

Question 1: Si l'un des tests échoue à cause d'un bug, comment pytest signale-t-il l'erreur et aide-t-il à la localiser ? Rédigez un test qui provoque volontairement une erreur, puis montrez la sortie du terminal obtenue.

Lorsqu'un test échoue, pytest montre quel test est faux.

Test mauvais:

```
def test_multiplication():
    my_calculator = Calculator()
    assert my_calculator.multiplication(5, 3) == 15
    assert my_calculator.multiplication(50, 0) == -1
    assert my_calculator.multiplication(50, -6) == -300
    assert my_calculator.multiplication(-10000, -1) == 10000
    assert my_calculator.multiplication(10000, 0.01) == 100
```

Resultat:

```
matias@vex027 MINGW64 ~/Downloads/LOG430/log430-labo0-e26 (main)
$ pytest
src\tests\test_calculator.py ...F. [100%]

===== FAILURES =====
_____ test_multiplication _____

  def test_multiplication():
      my_calculator = Calculator()
      assert my_calculator.multiplication(5, 3) == 15
>     assert my_calculator.multiplication(50, 0) == -1
E       assert 0 == -1
E       + where 0 = multiplication(50, 0)
E       +   where multiplication = <calculator.Calculator object at 0x000002A525FE8050>.multiplication

src\tests\test_calculator.py:34: AssertionError

===== short test summary info =====
FAILED src\tests\test_calculator.py::test_multiplication - assert 0 == -1
===== 1 failed, 4 passed in 0.09s =====
```

Question 2: Que fait GitHub pendant les étapes de « setup » et « checkout » ? Veuillez inclure la sortie du terminal GitHub CI dans votre réponse.

Setup fait la mise en place pour que l'image soit en mesure de faire toutes les tâches qui lui sont attendues.

Checkout fait la mise à jour du dépôt en effectuant un pull sur la version la plus récente.

```

build
succeeded 2 hours ago in 10s

Set up job 1s
1 Current runner version: '2.334.0'
2 ▶ Runner Image Provisioner
9 ▶ Operating System
13 ▶ Runner Image
18 ▶ GITHUB_TOKEN Permissions
22 Secret source: Actions
23 Prepare workflow directory
24 Prepare all required actions
25 Getting action download info
26 Download action repository 'actions/checkout@v3' (SHA:f43a0e5ff2bd294095638e18286ca9a3d1956744)
27 Download action repository 'actions/setup-python@v4' (SHA:7f4fc3e22c37d6ff65e88745f38bd3157c663f7c)
28 Complete job name: build

Checkout dépôt 1s
1 ▶ Run actions/checkout@v3
14 Syncing repository: Matis-Frenette-ETS/log430-labo0-e26
15 ▶ Getting Git version info
19 Temporarily overriding HOME='/home/runner/work/_temp/3b3f0673-7312-4e56-bdc1-a91424f88419' before making global git config changes
20 Adding repository directory to the temporary git global config as a safe directory
21 /usr/bin/git config --global --add safe.directory /home/runner/work/log430-labo0-e26/log430-labo0-e26
22 Deleting the contents of '/home/runner/work/log430-labo0-e26/log430-labo0-e26'
23 ▶ Initializing the repository
40 ▶ Disabling automatic garbage collection
42 ▶ Setting up auth
48 ▶ Fetching the repository
119 ▶ Determining the checkout info
120 ▶ Checking out the ref
124 /usr/bin/git log -1 --format='%H'
125 'e7be893e99bc94ab302b07c9d9682e470636a218'

```

Question 3: Quel type d'informations pouvez-vous obtenir via la commande top ? Veuillez donner quelques exemples. Veuillez inclure la sortie du terminal dans votre réponse.

La commande top retourne des informations sur l'utilisation de la machine, comme la memoire utiliser et les programmes qui sont entrain de fonctionner.

```

root@vm-matis-log430: ~
root@vm-matis-log430:~# top
top - 21:16:37 up 1:04, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 100 total, 1 running, 99 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3587.0 total, 120.7 free, 255.5 used, 3210.8 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3264.8 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 15100 root        39   19     0      0      0 S   0.3   0.0   0:00.03 arc_reap
 16001 root        20    0     0      0      0 I   0.3   0.0   0:00.06 kworker/0:1-events
    1 root        20    0 167384 12868  8320 S   0.0   0.4   0:09.05 systemd
    2 root        20    0     0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root         0 -20     0      0      0 I   0.0   0.0   0:00.00 rcu_gp
    4 root         0 -20     0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
    5 root         0 -20     0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
    6 root         0 -20     0      0      0 I   0.0   0.0   0:00.00 netns
    8 root         0 -20     0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
   10 root         0 -20     0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
   11 root        20    0     0      0      0 S   0.0   0.0   0:00.00 rcu_tasks_trace
   12 root        20    0     0      0      0 S   0.0   0.0   0:00.37 ksoftirqd/0
   13 root        20    0     0      0      0 I   0.0   0.0   0:01.27 rcu_sched
   14 root        rt     0     0      0      0 S   0.0   0.0   0:00.00 migration/0
   15 root        20    0     0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
   16 root        20    0     0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
   17 root         0 -20     0      0      0 I   0.0   0.0   0:00.00 inet_frag_wq
   18 root        20    0     0      0      0 S   0.0   0.0   0:00.00 kauditd

top - 21:16:41 up 1:04, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 100 total, 1 running, 99 sleeping, 0 stopped, 0 zombie
top - 21:16:41 up 1:04, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 100 total, 1 running, 99 sleeping, 0 stopped, 0 zombie

```